

Viannee J. Rosa-Arroyo

✉ vianneerosaarroyo@gmail.com  <https://www.linkedin.com/in/vianneerosa-arroyo>  github.com/vianneej

EDUCATION

Bachelor of Science in Computer Engineering

Expected May 2028 | Gainesville, FL

University of Florida, Herbert Wertheim College of Engineering

Relevant Coursework: Digital Logic, Data Structures & Algorithms, Microprocessor Applications, Introduction to Software Engineering

GPA: 3.2/4.0 | Certificates: Artificial Intelligence Fundamentals and Applications

PROFESSIONAL EXPERIENCE

UI/UX Director, Society of Hispanic Professional Engineers (SHPE)

08/2025 – Present

- Design interactive Figma prototypes, page layouts, and user flows for the SHPE website, improving usability and visual consistency across member-facing interfaces.
- Collaborate with iOS and Android developers to translate UI/UX designs into functional website updates while maintaining responsive and accessible design.
- Contributed to a Liquid Glass-inspired redesign across multiple interfaces, including the calendar and admin pages, helping standardize visual components, layouts, and styling.

PROJECTS

Horse Sanctuary Web Application | Code for Change Hackathon

02/2026 – 04/2026

Python | MongoDB Atlas | HTML | JavaScript | Render

- Built the frontend and contributed to backend development of a full-stack management platform, integrating the user interface with Python and MongoDB Atlas.
- Designed page layouts, color schemes, and individual horse profile and documentation pages for organizing care records, medications, medical information, and daily needs.
- Developed a system currently used by 5–7 core staff members to manage records for 70+ horses, replacing the sanctuary's paper-based recordkeeping process.
- Won MLH Best Use of MongoDB Atlas at Code for Change 2026, a 90-participant hackathon, for the platform developed for Retirement Home for Horses, Inc.

FRITA ESP32 Rover | Women in Computer Science and Engineering (WiCSE)

08/2025 – 04/2026

C++ | Python | ESP32 | Freenove 4WD Car Kit

- Assembled and integrated a Freenove 4WD ESP32 rover, connecting the chassis, motors, sensors, camera, and supporting hardware into a functional mobile platform.
- Programmed vehicle movement and control behavior, enabling directional control of the rover through the ESP32-based motor system.
- Helped develop a laptop-based control interface for remotely operating FRITA, integrating user controls with the rover's movement functions.

Motion In The Ocean | WiNGHacks, HTML | JavaScript | Digital Art

02/2026 – 02/2026

- Built JavaScript-driven interactivity and game logic, connecting the study timer and task tracker to in-game rewards and creature interactions.
- Created the application's original hand-drawn digital sea creatures and helped design the aquarium's visual style, animations, and interactive experience.
- Won Best Game at WiNGHacks 2026, a 99-participant hackathon, for developing a gamified productivity app that combines study tools with an interactive virtual aquarium.

Zoomi, C++ | Circuitry | Onshape | Arduino

01/2025 – 04/2025

- Built the circuitry using Arduino hardware for an automated cat toy, integrating electronics with a custom 3D-printed frame.
- Programmed motor movement and LED behavior in C++ using the Arduino IDE.
- Resolved a power-delivery limitation by implementing a dual 9V battery setup, enabling simultaneous operation of the wheels and electronic features.

TECHNICAL SKILLS

Programming Languages: C++, JavaScript, Python, HTML, VHDL, AVR Assembly

Hardware & Embedded: Arduino, uPad 2.0, AVR XMEGA, DE10-Lite, ESP32, Circuit Design

Tools & Platform: MongoDB, Git, Render, Microchip Studio, Quartus Prime, Digilent Analog Discovery, Waveforms, React, Node.js, Vercel, Firebase, Unity, Github Copilot, Claude

Design: Onshape, Figma, Procreate